1、 选取 old_degs.csv 或者 young_degs.csv 文件中提供的任意单个基因进行 GO 和 KEGG 数据库注释；

我选取的是 old_degs.csv 的 PGF 这个基因。

首先我在 NCBI 中搜索了 old_degs.csv 的 PGF，了解它的初步信息。

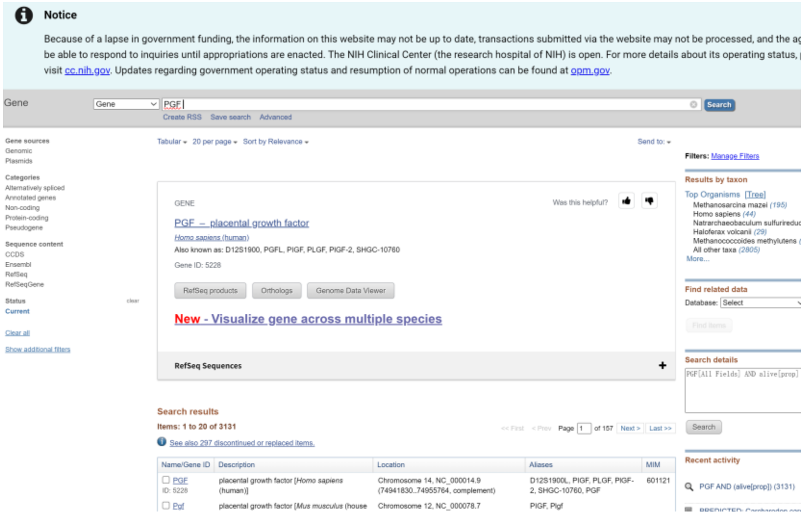

图 1-NCBI 搜索这个基因

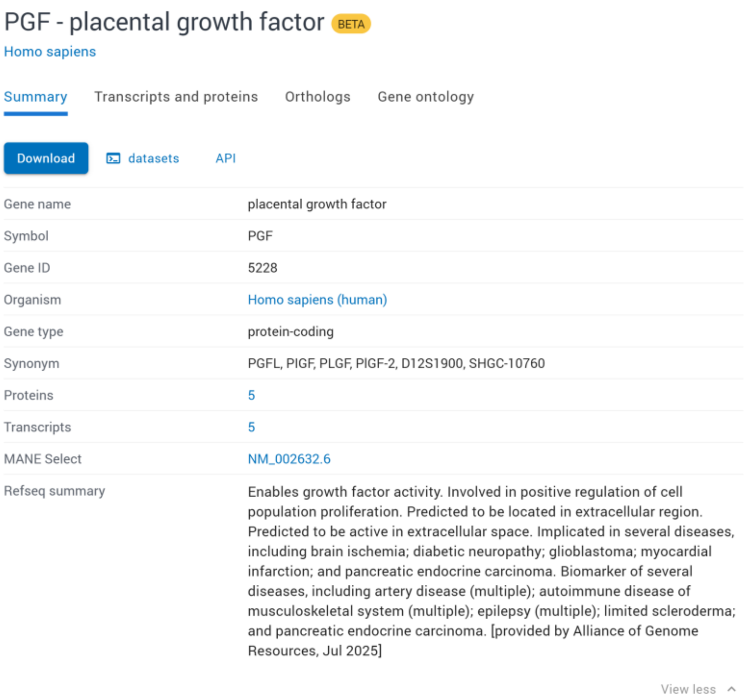

图 2-PGF 的详细说明

该基因的基本信息如图 2 所示，其为胎盘生长因子。在 QuickGO 数据库中查询 PGF 基因的 GO 注释，共获得 18 个 GO 注释，包括分子功能(F)等类别，显示该基因参与血管内皮生长因子受体结合、 生长因子活性等功能（图 3）。在图 3 中还可以看到 GO 注释的三个方面都有涉及，molecular fuction（绿），biological process(橙)，cellular component(紫)。

GO annotations

Taxon | Gene Products | GO terms | References | Aspect | Evidence | Extension | More | Clear all

Annotations | Statistics

Customise | Export | 18 annotations

Gene Product	Symbol	Qualifier	GO Term	Evidence	Reference	With / From	Taxon	Assigned By	Ann
UniProtKB:A0A096NQB0	PGF	enables	GO:0005172 F vascular endothelial growth factor receptor binding	ECO:0007826 IEA	GO_REF:0000118	PANTHER:PTN000246408	9555 Papio anubis	TreeGrafter	
UniProtKB:A0A096NQB0	PGF	enables	GO:0008083 F growth factor activity	ECO:0000501 IEA	GO_REF:0000120	UniProtKB:P49763 more...	9555 Papio anubis	UniProt	
UniProtKB:A0A096NQB0	PGF	enables	GO:0042056 F chemoattractant activity	ECO:0007826 IEA	GO_REF:0000118	PANTHER:PTN000246408	9555 Papio anubis	TreeGrafter	
UniProtKB:A0A096NQB0	PGF	enables	GO:0048018 F receptor ligand activity	ECO:0000256 IEA	GO_REF:0000117	ARBA:ARBA00028558	9555 Papio anubis	UniProt	
UniProtKB:A0A096NQB0	PGF	involved_in	GO:0001666 P response to hypoxia	ECO:0007826 IEA	GO_REF:0000118	PANTHER:PTN000246408	9555 Papio anubis	TreeGrafter	
UniProtKB:A0A096NQB0	PGF	involved_in	GO:0001938 P positive regulation of endothelial cell proliferation	ECO:0007826 IEA	GO_REF:0000118	PANTHER:PTN000246408	9555 Papio anubis	TreeGrafter	
UniProtKB:A0A096NQB0	PGF	involved_in	GO:0002040 P sprouting angiogenesis	ECO:0007826 IEA	GO_REF:0000118	PANTHER:PTN000246408	9555 Papio anubis	TreeGrafter	
UniProtKB:A0A096NQB0	PGF	involved_in	GO:0008284 P positive regulation of cell population proliferation	ECO:0000501 IEA	GO_REF:0000120	ARBA:ARBA00027538 more...	9555 Papio anubis	UniProt	
UniProtKB:A0A096NQB0	PGF	involved_in	GO:0038084 P vascular endothelial growth factor signaling pathway	ECO:0007826 IEA	GO_REF:0000118	PANTHER:PTN000246408	9555 Papio anubis	TreeGrafter	
UniProtKB:A0A096NQB0	PGF	involved_in	GO:0045766 P positive regulation of angiogenesis	ECO:0007826 IEA	GO_REF:0000118	PANTHER:PTN000246408	9555 Papio anubis	TreeGrafter	
UniProtKB:A0A096NQB0	PGF	involved_in	GO:0048010 P vascular endothelial growth factor receptor signaling pathway	ECO:0007826 IEA	GO_REF:0000118	PANTHER:PTN000246408	9555 Papio anubis	TreeGrafter	
UniProtKB:A0A096NQB0	PGF	involved_in	GO:0050918 P positive chemotaxis	ECO:0000366 IEA	GO_REF:0000108	GO:0042056	9555 Papio anubis	GOC	
UniProtKB:A0A096NQB0	PGF	involved_in	GO:0050930 P induction of positive chemotaxis	ECO:0007826 IEA	GO_REF:0000118	PANTHER:PTN000246408	9555 Papio anubis	TreeGrafter	
UniProtKB:A0A096NQB0	PGF	involved_in	GO:0051781 P positive regulation of cell division	ECO:0007322 IEA	GO_REF:0000043	UniProtKB-KW:KW-0497	9555 Papio anubis	UniProt	
UniProtKB:A0A096NQB0	PGF	involved_in	GO:0060754 P positive regulation of mast cell chemotaxis	ECO:0007826 IEA	GO_REF:0000118	PANTHER:PTN000246408	9555 Papio anubis	TreeGrafter	
UniProtKB:A0A096NQB0	PGF	located_in	GO:0005576 C extracellular region	ECO:0007322 IEA	GO_REF:0000044	UniProtKB-SubCell:SL-0243	9555 Papio anubis	UniProt	
UniProtKB:A0A096NQB0	PGF	located_in	GO:0005615 C extracellular space	ECO:0007826 IEA	GO_REF:0000118	PANTHER:PTN000246408	9555 Papio anubis	TreeGrafter	
UniProtKB:A0A096NQB0	PGF	located_in	GO:0016020 C membrane	ECO:0000256 IEA	GO_REF:0000002	InterPro:IPR000072	9555 Papio anubis	InterPro	

图 3

Search KEGG **for** PGF Go Clear

Database: KEGG - Search term: PGF

KEGG ORTHOLOGY

K16859
PGF; placenta growth factor
K23486
PGFS; 9,11-endoperoxide prostaglandin H2 reductase [EC:1.1.1.-]

KEGG GENES

hsa:5228
K16859 placenta growth factor | (RefSeq) PGF, D12S1900, PGFL, PIGF, PLGF, PlGF-2, SHGC-10760; placental growth factor
hsa:10734
K13055 cohesin complex subunit SA-3 | (RefSeq) STAG3, SA3, SPGF61; STAG3 cohesin complex component
hsa:145645
K25750 telomere repeats-binding bouquet formation protein 2 | (RefSeq) TERB2, C15orf43, SPGF59; telomere repeat binding bouquet formation protein 2
hsa:140801
K02866 large subunit ribosomal protein L10e | (RefSeq) RPL10L, RPL10_5_1358, SPGF63; ribosomal protein L10 like
hsa:11055
K25752 zona pellucida-binding protein | (RefSeq) ZPBP, SPGF66, ZPBP1; zona pellucida binding protein
• • • » display all

KEGG GENOME

T08191
pgf; Pseudomonas germanica FIT28

图 4

Homo sapiens (human): 5228

Help

Entry	5228 CDS T01001
Symbol	PGF, D12S1900, PGFL, PIGF, PLGF, PlGF-2, SHGC-10760
Name	(RefSeq) placental growth factor
KO	K16859 placenta growth factor
Organism	hsa Homo sapiens (human)
Pathway	hsa04010 MAPK signaling pathway hsa04014 Ras signaling pathway hsa04015 Rap1 signaling pathway hsa04151 PI3K-Akt signaling pathway hsa04510 Focal adhesion hsa05200 Pathways in cancer
Network	nt06526 MAPK signaling nt06530 PI3K signaling
Element	N01592 GF-RTK-RAS-ERK signaling pathway N01656 GF-RTK-PI3K signaling pathway N01658 GF-RTK-RAS-PI3K signaling pathway
Drug target	Aflibercept: D09574<JP/US> Aflibercept beta: D10819<JP>
Brite	KEGG Orthology (KO) [BR:hsa00001] 09130 Environmental Information Processing 09132 Signal transduction 04010 MAPK signaling pathway 5228 (PGF) 04014 Ras signaling pathway 5228 (PGF) 04015 Rap1 signaling pathway 5228 (PGF) 04151 PI3K-Akt signaling pathway 5228 (PGF) 09140 Cellular Processes 09144 Cellular community - eukaryotes 04510 Focal adhesion 5228 (PGF) 09160 Human Diseases 09161 Cancer: overview 05200 Pathways in cancer 5228 (PGF) 09180 Brite Hierarchies 09183 Protein families: signaling and cellular processes 04052 Cytokines and neuropeptides [BR:hsa04052] 5228 (PGF) Cytokines and neuropeptides [BR:hsa04052] Cytokines Growth factors (RTK binding) 5228 (PGF) BRITE hierarchy
SSDB	Ortholog Paralog Gene cluster GFIT
Motif	Pfam: PDGF Motif
Other DBs	NCBI-GeneID: 5228 NCBI-ProteinID: NP_002623 OMIM: 601121 HGNC: 8893 Ensembl: ENSG00000119630 UniProt: P49763 Q53XY6 Q86TW6
Structure	PDB PDBj
Position	14:complement(74941830..74955764) Genome browser
AA seq	170 aa AA seq DB search MPVMRLFPCFLQLLAGLALPAVPPQQWALSAGNGSSEVEVVPFQEVWGRSYCRALERLVD VVSEYPSEVEHMFSPSCVSLLRCTGCCGDENLHCVPVETANVTMQLLKIRSGDRPSYVEL TFSQHVRCECRPLREKMKPERRRPKGRGKRRREKQRPTDCHLCGDAVPRR
NT seq	513 nt NT seq atgccggtcatgaggctgttcccttgcttcctgcagctcctggccgggctggcgctgcct gctgtgcccccccagcagtgggccttgtctgctgggaacggctcgtcagaggtggaagtg gtacccttccaggaagtgtggggccgcagctactgccgggcgctggagaggctggtggac gtcgtgtccgagtaccccagcgaggtggagcacatgttcagcccatcctgtgtctccctg ctgcgctgcaccggctgctgcggcgatgagaatctgcactgtgtgccggtggagacggcc aatgtcaccatgcagctcctaaagatccgttctggggaccggccctcctacgtggagctg acgttctctcagcacgttcgctgcgaatgccggcctctgcgggagaagatgaagccggaa aggaggagacccaagggcaggggaagaggaggagagagaagcagagacccacagactgc cacctgtgcggcgatgctgttccccggaggtaa

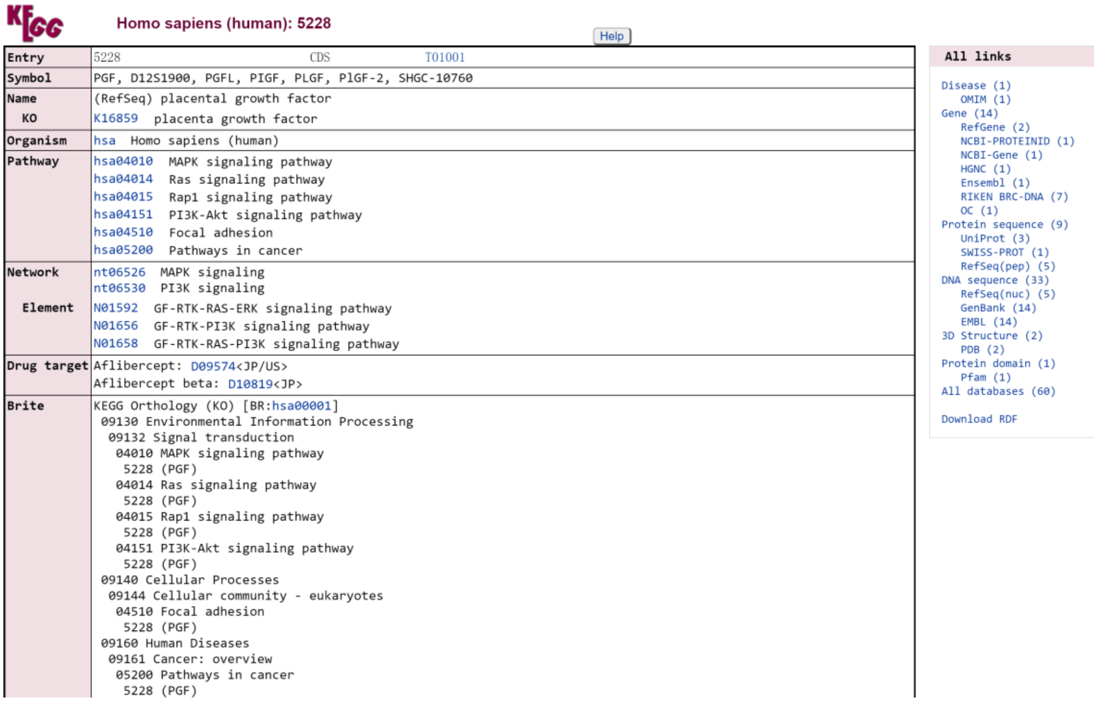

图 5-

KEGG 结果如上，主要涉及的通路如图 6，具体为：

1. hsa04010 - MAPK signaling pathway（信号转导）
2. hsa04014 - Ras signaling pathway（信号转导）
3. hsa04015 - Rap1 signaling pathway（信号转导）
4. hsa04151 - PI3K-Akt signaling pathway（信号转导）
5. hsa04510 - Focal adhesion（细胞通讯）
6. hsa05200 - Pathways in cancer（人类疾病-癌症）

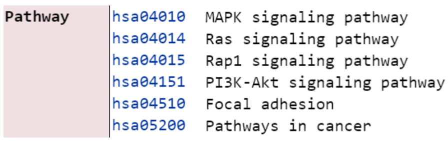

图 6-所有涉及的通路

其中点入任意一个 Pathway，会显示如下图所示的详细的的通路图，比如说我就点进去第一个 hsa0410 查看了 MAPK 的通路（图 7）。其他的通路我不一一列出。

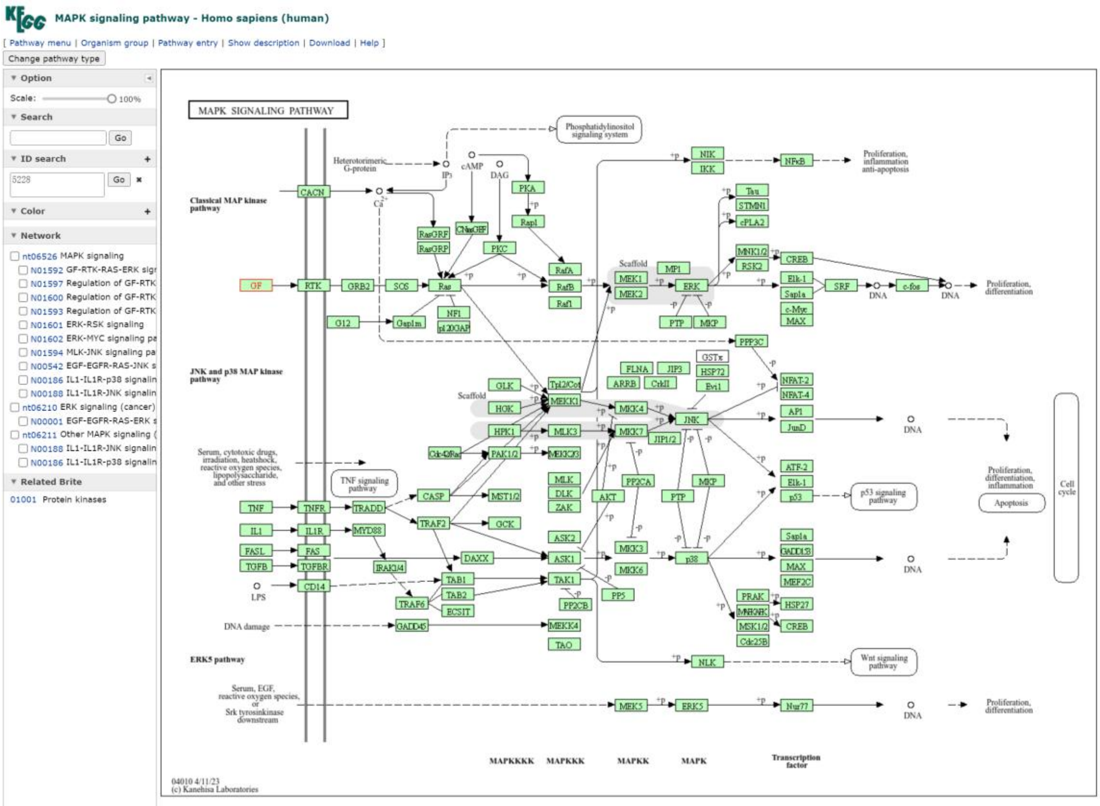

图 7

这 6 条重要的 KEGG 信号通路通路集中在：

（1）信号转导 MAPK、Ras、Rap1、PI3K-Akt 通路；提示 PGF 在细胞信号传递中的核心作用

（2）细胞通讯：Focal adhesion（粘着斑）；提示 PGF 参与细胞-细胞外基质相互作用

（3）疾病相关：癌症通路 提示 PGF 在肿瘤血管生成中的重要作用；

这些通路与血管生成、细胞增殖、肿瘤发生发展密切相关，解释了 PGF 在老年相关血管疾病和 肿瘤中的重要作用。

2、选取部分基因集合使用 DAVID、或者 clusterProfiler 或者 Metascape 进行富集分析；

我使用 clusterProfiler 完成这项任务。

通过 GO 和 KEGG 对衰老组中 402 个基因进行富集分析，发现老年组差异基因主要富集在肌肉发育、心脏功能、细胞粘附和信号转导相关的生物学过程和通路。这些发现揭示了，肌肉组织相关基因的系统性改变可能是老年肌肉减少的分子基础，心脏功能和钙稳态相关通路的变化提示心血管系统的衰老变化，细胞粘附和细胞外基质的改变反映了组织结构和功能的衰退。多条信号通路的协同变化说明衰老是一个系统性的生物学过程。

我的富集分析是合理的、有效的，因为一个好的富集分析的功能之间是要有生物学相关性的，肌肉组织与心脏功能调控之间，就具有相关性；且这与老年性肌肉减少和心血管功能下降的生理现象一致。所以，这项富集分析研究，能为理解衰老的分子机制提供重要线索。

```
> print(paste("成功转换", nrow(gene_entrez), "个基因"))
[1] "成功转换 402 个基因"
>
```

图 8-我所使用的表格中一共需要富集的基因数量是 402 个基因

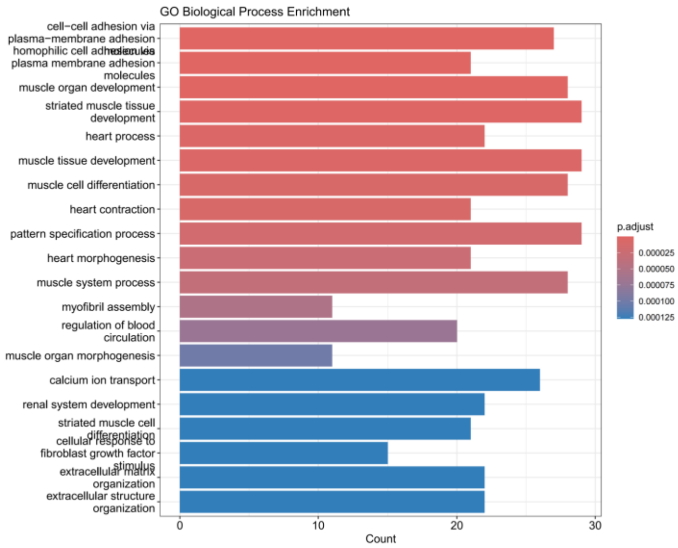

图 9-GO-BP 富集结果（柱状图）

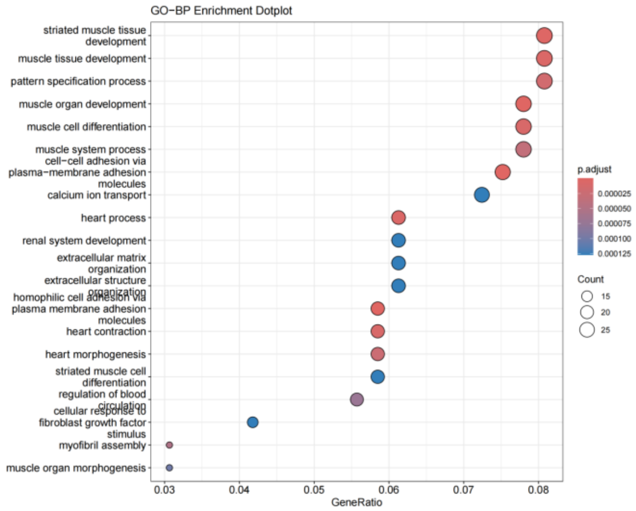

图 10-GO-BP 富集结果（气泡图—富集程度与显著性）

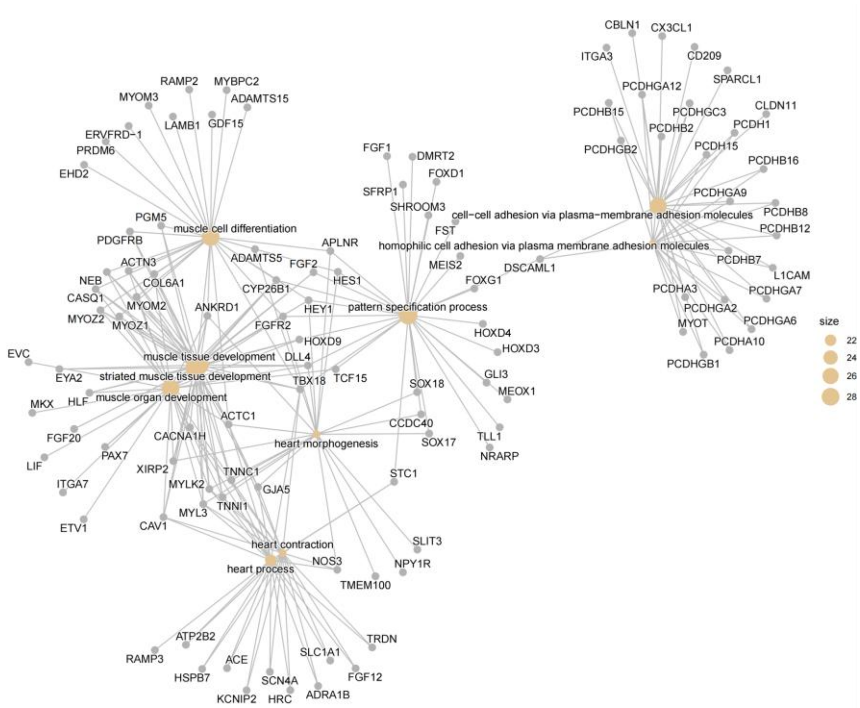

图 10-GO-BP 富集结果（网络图—呈现基因和功能的关系）

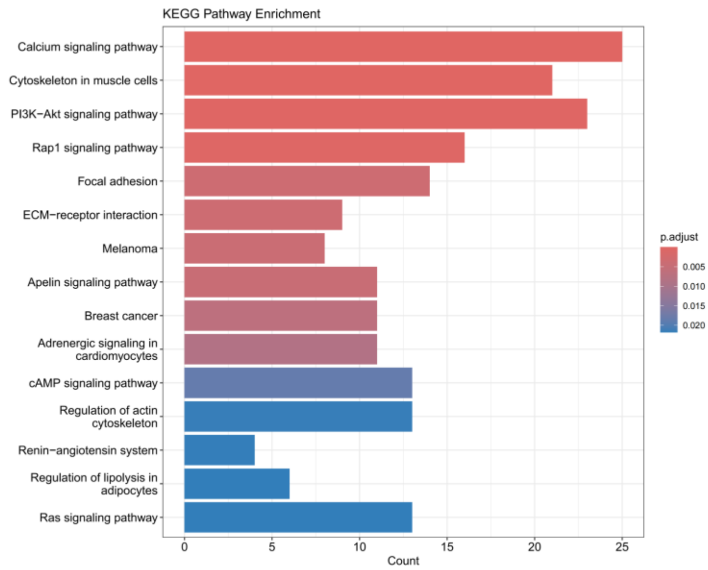

图 11-KEGG 富集结果（柱形图）

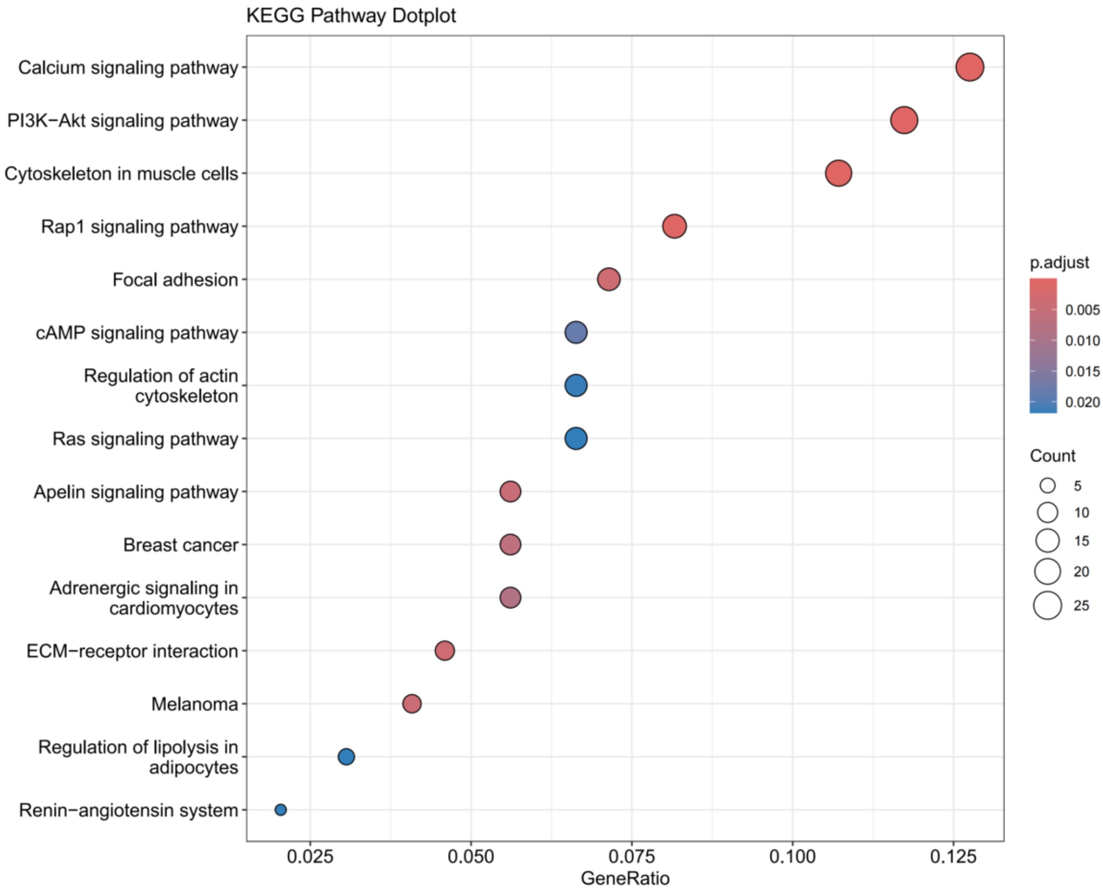

图 12-KEGG 富集结果（气泡图）

相关代码如下（见下页）：

```r
# 1. 加载包
library(clusterProfiler)
library(org.Hs.eg.db)
library(enrichplot)
library(ggplot2)

# 2. 读取基因列表
old_genes <- read.csv("old_degs.csv")$GeneID
print(paste("读取到", length(old_genes), "个基因"))

# 3. 转换基因 ID
gene_entrez <- bitr(old_genes,
                    fromType = "SYMBOL",
                    toType = "ENTREZID",
                    OrgDb = org.Hs.eg.db)

print(paste("成功转换", nrow(gene_entrez), "个基因"))

# 4. GO 富集分析
go_bp <- enrichGO(gene = gene_entrez$ENTREZID,
                  OrgDb = org.Hs.eg.db,
                  ont = "BP",
                  pAdjustMethod = "BH",
                  pvalueCutoff = 0.05,
                  qvalueCutoff = 0.2,
                  readable = TRUE)

print("GO-BP 富集结果:")
print(head(go_bp@result, 10))

# 5. KEGG 富集分析
kegg <- enrichKEGG(gene = gene_entrez$ENTREZID,
                   organism = "hsa",
                   pvalueCutoff = 0.05,
                   qvalueCutoff = 0.2)

kegg <- setReadable(kegg, OrgDb = org.Hs.eg.db, keyType =
"ENTREZID")

print("KEGG 富集结果:")
print(head(kegg@result, 10))

# 6. 可视化 - 柱状图
p1 <- barplot(go_bp,
              showCategory = 20,
              title = "GO Biological Process Enrichment")
```

```r
print(p1)
ggsave("GO_BP_barplot.pdf", p1, width = 10, height = 8)

# 7. 可视化 - 气泡图
p2 <- dotplot(go_bp,
              showCategory = 20,
              title = "GO-BP Enrichment Dotplot")
print(p2)
ggsave("GO_BP_dotplot.pdf", p2, width = 10, height = 8)

# 8. 可视化 - 网络图
p3 <- cnetplot(go_bp,
               showCategory = 10,
               cex_label_category = 0.8)
print(p3)
ggsave("GO_BP_network.pdf", p3, width = 12, height = 10)

# 9. KEGG 可视化
p4 <- barplot(kegg,
              showCategory = 15,
              title = "KEGG Pathway Enrichment")
print(p4)
ggsave("KEGG_barplot.pdf", p4, width = 10, height = 8)

p5 <- dotplot(kegg,
              showCategory = 15,
              title = "KEGG Pathway Dotplot")
print(p5)
ggsave("KEGG_dotplot.pdf", p5, width = 10, height = 8)

# 10. 保存结果
write.csv(go_bp@result, "GO_BP_enrichment_results.csv", row.names = FALSE)
write.csv(kegg@result, "KEGG_enrichment_results.csv", row.names = FALSE)
```